

Prime-return bouquet height and analytic radius: midpoint compression and prime-order anchors

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August 2026

Abstract

For the integral Hénon automorphism $H(x, y) = (1 - 6x^2 - y, x)$ and the reversing-axis seed $P_0 = (0, 0)$, let r_p be the return period of P_0 modulo a rational prime p . RH-362 attached one complete marked r_p -cycle to each prime and obtained a locally finite formal Artin–Mazur product. We prove that this formal object has a nontrivial analytic domain.

Reversibility compresses every two-coordinate gcd term to one midpoint coordinate difference: $a_{2k} = |x_k - x_{k-1}|$ and $a_{2k+1} = |x_{k+1} - x_{k-1}|$. The escaping negative orbit then satisfies $5b_n^2 \leq b_{n+1} \leq 6b_n^2$, which yields explicit two-sided bounds and $\log a_n = \Theta(2^{n/2})$. For the bouquet fixed-point ledger $T_n = \sum_{p:r_p|n} r_p$, we obtain the all-order envelope

$$T_n \leq (\log_2 30)n 2^{\lceil n/2 \rceil - 2}.$$

Consequently $\prod_p (1 - z^{r_p})^{-1}$ converges normally and is holomorphic and zero-free on $|z| < 2^{-1/2}$.

At every odd prime order ℓ , the term a_ℓ has a primitive prime divisor and $[z^\ell] \log \mathcal{Z}_0 = \omega(a_\ell) \geq 1$. This is an Euler-exponent and logarithmic-coefficient anchor, not a raw zeta coefficient: decomposable lower-order cycles already change the coefficient at order seven. Finally, the naive Hilbert direct sum $\bigoplus_p zU_p$ is noncompact and belongs to no finite Schatten class for $z \neq 0$; therefore the analytic Euler product is not its ordinary Fredholm determinant.

The bouquet remains a marked disjoint union across distinct finite fields, not a full finite-field zeta, Hasse–Weil factor, canonical global Hénon operator, or von-Mangoldt trace model. Gates A–E remain false/open, and no Hilbert–Pólya, Riemann-zero, completed-divisor, or RH conclusion is claimed.

1 Frozen foundation and exact object

The RH-1–RH-361 foundation remains frozen in four provenance-preserving volumes [5]. RH-362 opened an independent arithmetic-dynamical branch by assigning one marked modular return cycle to each prime [3]; RH-364 is the immediate numbered baseline and leaves the original physical route coordinate

$$\text{actual_same_clock_unnormalized_head_transport_open} \tag{1.1}$$

unchanged [4]. Nothing below estimates that physical same-clock defect, supplies a typed q/E_{off} theorem, or closes the RH-241 moving noisy envelope.

Fix the integral polynomial automorphism

$$H(x, y) = (1 - 6x^2 - y, x), \quad H^{-1}(u, v) = (v, 1 - 6v^2 - u). \tag{1.2}$$

Let

$$R(x, y) = (y, x). \tag{1.3}$$

Direct substitution gives

$$R^2 = \text{id}, \quad RHR = H^{-1}. \tag{1.4}$$

These identities hold over every commutative ring and hence after reduction modulo every positive integer [6].

For $P \in \mathbb{Z}^2$, let

$$r_M(P) = \min\{n \geq 1 : H^n(P) \equiv P \pmod{M}\}, \quad r_1(P) = 1, \quad (1.5)$$

and, writing $H^n(P) = (x_n, y_n)$, define

$$a_n(P) = \gcd(|x_n - x_0|, |y_n - y_0|). \quad (1.6)$$

The locked source theorem is

$$M \mid a_n(P) \iff r_M(P) \mid n. \quad (1.7)$$

It also proves strong divisibility for every nonperiodic integral orbit [1, 6].

This paper uses only

$$P_0 = (0, 0). \quad (1.8)$$

Set $x_{-1} = x_0 = 0$ and define

$$x_{n+1} = 1 - 6x_n^2 - x_{n-1}. \quad (1.9)$$

Then

$$H^n(P_0) = (x_n, x_{n-1}), \quad a_n := a_n(P_0) = \gcd(|x_n|, |x_{n-1}|). \quad (1.10)$$

2 Exact reversibility midpoint compression

The seed lies on the reversing axis: $RP_0 = P_0$. Thus

$$H^{-k}P_0 = RH^kP_0 \quad (k \geq 0). \quad (2.1)$$

This turns a long return into equality of two midpoint states.

Theorem 2.1 (Midpoint identities). *For every $k \geq 1$,*

$$a_{2k} = |x_k - x_{k-1}|, \quad (2.2)$$

and for every $k \geq 0$,

$$a_{2k+1} = |x_{k+1} - x_{k-1}|. \quad (2.3)$$

Proof. Fix a positive modulus M . Because H is invertible modulo M ,

$$\begin{aligned} H^{2k}P_0 \equiv P_0 \pmod{M} &\iff H^kP_0 \equiv H^{-k}P_0 \pmod{M} \\ &\iff (x_k, x_{k-1}) \equiv (x_{k-1}, x_k) \pmod{M} \\ &\iff M \mid x_k - x_{k-1}. \end{aligned}$$

Together with (1.7), this says that the two nonnegative integers in (2.2) have exactly the same positive divisors, hence are equal.

Likewise,

$$\begin{aligned} H^{2k+1}P_0 \equiv P_0 \pmod{M} &\iff H^{k+1}P_0 \equiv H^{-k}P_0 \pmod{M} \\ &\iff (x_{k+1}, x_k) \equiv (x_{k-1}, x_k) \pmod{M} \\ &\iff M \mid x_{k+1} - x_{k-1}, \end{aligned}$$

which proves (2.3). The convention $x_{-1} = 0$ includes $k = 0$. □

Remark 2.2. The theorem is specific to the reversing-axis seed P_0 . It is not a height theorem for every integral point and is not inferred from the finite return-rank tables.

3 Quadratic escape and the two-sided height scale

The first scalar values are

$$x_0 = 0, \quad x_1 = 1, \quad x_2 = -5, \quad x_3 = -150. \quad (3.1)$$

For $n \geq 2$, put

$$b_n = -x_n. \quad (3.2)$$

Theorem 3.1 (Quadratic height recursion and closed bounds). *The sequence b_n is positive and strictly increasing. It satisfies*

$$b_2 = 5, \quad b_3 = 150 = 6b_2^2, \quad (3.3)$$

and, for $n \geq 3$,

$$b_{n+1} = 6b_n^2 - b_{n-1} - 1. \quad (3.4)$$

For every $n \geq 2$,

$$5b_n^2 \leq b_{n+1} \leq 6b_n^2. \quad (3.5)$$

Consequently,

$$5^{2^{n-1}-1} \leq b_n \leq \frac{30^{2^{n-2}}}{6}. \quad (3.6)$$

Proof. The bases follow from (1.9). Suppose $b_n > b_{n-1} \geq 5$. Because the variables are integral, $b_{n-1} + 1 \leq b_n$. Equation (1.9) gives (3.4), and hence

$$b_{n+1} \geq 6b_n^2 - b_n > b_n.$$

This proves positivity and strict increase by induction from $(b_2, b_3) = (5, 150)$. The upper inequality in (3.5) is immediate from (3.4); for the lower inequality,

$$b_{n-1} + 1 \leq b_n \leq b_n^2$$

gives $b_{n+1} \geq 5b_n^2$. The case $n = 2$ follows directly from (3.3).

Multiplying the lower recurrence by five gives $5b_{n+1} \geq (5b_n)^2$, and multiplying the upper recurrence by six gives $6b_{n+1} \leq (6b_n)^2$. Iterating from $5b_2 = 25$ and $6b_2 = 30$ yields (3.6). $\square$

Strict increase proves again that the integral orbit of P_0 is nonperiodic. In particular every a_n is positive.

Theorem 3.2 (Explicit two-sided bounds for the gcd sequence). *The first values are*

$$a_1 = a_2 = 1, \quad a_3 = 5, \quad a_4 = 6, \quad a_5 = 151. \quad (3.7)$$

Let $n \geq 6$ and $m = \lceil n/2 \rceil$. Then

$$\frac{24}{25}b_m \leq a_n < b_m. \quad (3.8)$$

Uniformly for every $n \geq 3$,

$$4 \cdot 5^{2^{m-1}-2} \leq a_n \leq \frac{30^{2^{m-2}}}{5}, \quad m = \left\lceil \frac{n}{2} \right\rceil. \quad (3.9)$$

Thus

$$\log a_n = \Theta(2^{n/2}). \quad (3.10)$$

Proof. The bases follow from theorem 2.1. If $n = 2m \geq 6$, then

$$a_n = b_m - b_{m-1}.$$

If $n = 2m - 1 \geq 7$, then

$$a_n = b_m - b_{m-2}.$$

In either case the subtracted term is at most b_{m-1} . Since

$$b_m \geq 5b_{m-1}^2 \geq 25b_{m-1},$$

equation (3.8) follows.

The three special values in (3.7), together with (3.8), give the uniform comparison

$$\frac{4}{5}b_m \leq a_n \leq \frac{6}{5}b_m \quad (n \geq 3). \quad (3.11)$$

Combining this with (3.6) proves (3.9). Finally 2^{m-1} is comparable to $2^{n/2}$, proving (3.10). $\square$

4 The all-order bouquet trace envelope

For a rational prime p , write $r_p = r_p(P_0)$, and let

$$\mathcal{O}_p = \{P_0, H_p P_0, \dots, H_p^{r_p-1} P_0\} \quad (4.1)$$

be the complete marked cycle in $\mathbb{F}_p^2$. Let U_p be its cyclic permutation matrix. Then

$$\text{Tr}(U_p^n) = r_p \mathbf{1}_{r_p|n}, \quad \det(I - zU_p) = 1 - z^{r_p}. \quad (4.2)$$

Define the countable marked disjoint union

$$\mathcal{X}_0 = \bigsqcup_p \mathcal{O}_p \quad (4.3)$$

and let F_0 act on each component as the cyclic restriction of H_p :

$$F_0|_{\mathcal{O}_p} = H_p|_{\mathcal{O}_p}. \quad (4.4)$$

Define also the rank multiplicities

$$c_d = \#\{p : r_p = d\}. \quad (4.5)$$

The return-divisibility theorem implies that c_d is finite for every d : every such prime divides the fixed positive integer a_d .

Theorem 4.1 (Exact fixed-point ledger and all-order envelope). *For every $n \geq 1$, the fixed-point set of the n -th bouquet iterate is finite and*

$$T_n := \# \text{Fix}(F_0^n) = \sum_{p: r_p|n} r_p = \sum_{d|n} d c_d = \sum_{p|a_n} r_p. \quad (4.6)$$

Moreover $T_1 = T_2 = 0$, and for every $n \geq 1$,

$$T_n \leq (\log_2 30)n 2^{\lceil n/2 \rceil - 2}. \quad (4.7)$$

Proof. Equation (4.2) and finiteness of the set $r_p \leq n$ give the first three expressions in (4.6). Equation (1.7) gives the last.

If $p \mid a_n$, then $r_p \mid n$, hence $r_p \leq n$. Therefore

$$T_n \leq n \omega(a_n), \quad (4.8)$$

where ω counts distinct prime factors. Since $2^{\omega(a_n)} \leq \text{rad}(a_n) \leq a_n$,

$$\omega(a_n) \leq \log_2 a_n. \quad (4.9)$$

For $n \geq 3$, theorem 3.2 gives

$$\log_2 a_n \leq 2^{\lceil n/2 \rceil - 2} \log_2 30 - \log_2 5,$$

which proves (4.7). The terms $a_1 = a_2 = 1$ give $T_1 = T_2 = 0$, so the same displayed envelope holds at the two base orders. $\square$

Remark 4.2 (Data type). The term *trace ledger* refers to the finite sum of local cycle traces, or equivalently the bouquet fixed-point count. It is not yet the Hilbert-space trace of a trace-class global operator. Section 7 proves the precise obstruction.

5 A strict zero-free disk

The RH-362 formal Artin–Mazur product is

$$\mathcal{Z}_0(z) = \exp \left(\sum_{n \geq 1} \frac{T_n}{n} z^n \right) = \prod_p (1 - z^{r_p})^{-1}. \quad (5.1)$$

RH-362 asserted this coefficientwise, without a positive analytic radius.

Theorem 5.1 (Normal convergence and zero-free disk). *For every*

$$0 \leq r < 2^{-1/2}, \quad (5.2)$$

one has

$$\sum_{n \geq 1} \frac{T_n}{n} r^n \leq (\log_2 30) \frac{r^3 + r^4}{1 - 2r^2}. \quad (5.3)$$

Consequently the logarithmic series and Euler product in (5.1) converge locally normally on

$$|z| < 2^{-1/2}. \quad (5.4)$$

They define a holomorphic zero-free function $\mathcal{Z}_0$ there, and $\mathcal{Z}_0^{-1}$ is also holomorphic and zero-free.

Proof. The base traces vanish. The odd and even parts of (4.7) give

$$\sum_{n \geq 1} \frac{T_n}{n} r^n \leq (\log_2 30) \left(\sum_{j \geq 0} 2^j r^{2j+3} + \sum_{j \geq 0} 2^j r^{2j+4} \right),$$

which is (5.3).

For $|z| \leq r < 2^{-1/2}$, Tonelli's theorem and (4.6) give

$$\sum_p \sum_{j \geq 1} \frac{|z|^{j r_p}}{j} = \sum_{n \geq 1} \frac{T_n}{n} |z|^n < \infty. \quad (5.5)$$

Thus the analytic logarithms $\sum_{j \geq 1} z^{j r_p} / j$ sum absolutely and uniformly on compact subdisks. Exponentiating their sum and its negative proves normal convergence and the two zero-free claims. $\square$

The inequality is strict at the boundary: the denominator in (5.3) vanishes when $r = 2^{-1/2}$. No assertion is made there.

Corollary 5.2 (Origin Taylor-radius bracket). *Let $R_{\log}$ and R_Z be the origin Taylor radii of $\log Z_0$ and Z_0 , respectively. Then*

$$2^{-1/2} \leq R_{\log} \leq 1, \quad 2^{-1/2} \leq R_Z \leq 1. \quad (5.6)$$

Proof. The lower bounds are theorem 5.1. The upper bounds follow from the prime-order anchors in theorem 6.2: the logarithmic coefficient is at least one at infinitely many orders, and the coefficients of the Euler product are nonnegative with the raw coefficient at least the primitive contribution. $\square$

This is a bracket for the origin power-series radii. It is not an exact radius, a meromorphic-continuation theorem, or a natural-boundary theorem.

6 Odd-prime primitive anchors and the raw-coefficient firewall

Definition 6.1. A prime p is a primitive prime divisor of a_n when $p \mid a_n$ and $p \nmid a_m$ for every $1 \leq m < n$.

Theorem 6.2 (Primitive anchors at every odd prime order). *Let ℓ be an odd rational prime. Then $a_\ell > 1$, and*

$$\{p : p \mid a_\ell\} = \{p : r_p = \ell\}. \quad (6.1)$$

Every distinct prime divisor of a_ℓ is primitive. In particular,

$$c_\ell = \omega(a_\ell) \geq 1, \quad T_\ell = \ell c_\ell, \quad [z^\ell] \log Z_0(z) = c_\ell. \quad (6.2)$$

The number of primitive bouquet cycles of length ℓ is exactly c_ℓ .

Proof. The lower bound in theorem 3.2 gives $a_\ell \geq 4$, so a_ℓ has a prime divisor. Equation (1.7) gives

$$p \mid a_\ell \iff r_p \mid \ell.$$

No prime has $r_p = 1$, because that would imply $p \mid a_1 = 1$. Since ℓ is prime, the only remaining divisor is $r_p = \ell$, proving (6.1). If such a p divided a_m for some $m < \ell$, then $\ell = r_p \mid m$, a contradiction. The first two identities in (6.2) follow.

Finally,

$$[z^\ell] \log Z_0 = \frac{T_\ell}{\ell} = c_\ell.$$

Equivalently, Möbius inversion of the fixed-point ledger gives primitive cycle count $(T_\ell - T_1)/\ell = c_\ell$. $\square$

The theorem does not assert that a_ℓ is squarefree. The symbol ω counts distinct factors, not factors with multiplicity. It also does not imply a primitive divisor at any composite order.

Proposition 6.3 (The anchor is not the raw zeta coefficient). *Write*

$$Z_0(z) = \sum_{n \geq 0} q_n z^n. \quad (6.3)$$

Then

$$nq_n = \sum_{k=1}^n T_k q_{n-k}, \quad q_0 = 1. \quad (6.4)$$

The exact finite ledger begins

$$(q_0, \dots, q_7) = (1, 0, 0, 1, 2, 1, 2, 3). \quad (6.5)$$

At the prime order seven,

$$c_7 = 1, \quad [z^7] \log \mathcal{Z}_0 = 1, \quad [z^7] \mathcal{Z}_0 = 3. \quad (6.6)$$

Proof. Differentiating (5.1) and comparing coefficients gives (6.4). The midpoint identities give

$$a_3 = 5, \quad a_4 = 6, \quad a_7 = 134989.$$

Direct exact factorization gives rank multiplicities $c_3 = 1, c_4 = 2, c_7 = 1$; the artifact independently checks the return ranks and the primality of 134989. Thus

$$(T_1, \dots, T_7) = (0, 0, 3, 8, 5, 9, 7).$$

Substitution in (6.4) proves (6.5) and (6.6). The two extra order-seven contributions are decomposable products of one length-three and one of the two length-four cycles. $\square$

7 Naive direct-sum obstruction

Let

$$\mathcal{H}_0 = \bigoplus_p \mathbb{C}^{r_p}, \quad U = \bigoplus_p U_p, \quad B_z = zU. \quad (7.1)$$

The operator U is unitary. Formally, its finite blocks have the local determinants used in (5.1). Ordinary Fredholm determinants, however, require trace-ideal hypotheses [2].

Theorem 7.1 (Noncompactness and the Fredholm firewall). *For every $z \neq 0$, the operator B_z is noncompact and belongs to no finite Schatten class S_q . For every $n \geq 1$, U^n is not trace class, so T_n is not its Hilbert-space trace. Consequently $\mathcal{Z}_0^{-1}$ is not the ordinary Fredholm determinant $\det_F(I - B_z)$ of this naive direct sum.*

For $|z| < 1$, the operator $I - B_z$ is nevertheless invertible. Noncompactness is not a non-Fredholmness statement.

Proof. Choose one unit vector e_p in each prime block. The vectors $B_z e_p$ are mutually orthogonal and all have norm $|z|$. They therefore have no convergent subsequence when $z \neq 0$, proving noncompactness.

Every singular value of U_p is one. Hence the partial S_q burden of B_z over any finite prime set S is

$$|z|^q \sum_{p \in S} r_p. \quad (7.2)$$

This diverges as S exhausts the primes, so no finite S_q membership holds. The same observation with $z = 1$ shows that every U^n has infinitely many singular values equal to one and is not trace class. Thus the finite block sum in (4.6) is a fixed-point ledger, not a global operator trace.

Finally, if $|z| < 1$, then $\|B_z\| = |z| < 1$, so the Neumann series $\sum_{j \geq 0} B_z^j$ inverts $I - B_z$. $\square$

This theorem excludes only the naive constant-weight cycle direct sum. It is not a nonexistence theorem for every compact realization of the same scalar germ.

8 Route verdicts and Gate ledger

Route	Exact verdict
Route A	GO. Reversibility supplies exact midpoint compression; escape gives the two-sided height scale; the resulting all-order fixed-point envelope produces a strict zero-free disk and infinitely many prime-order primitive anchors. The direct-sum theorem supplies a rigorous scoped operator negative.
Route B	STOP_SCOPED. The first fatal mismatch is Gate A data type: one marked cycle is selected from each of distinct finite-field maps, so the bouquet is not a canonical intrinsic global Hénon operator. Its fixed-point counts are also not signed von-Mangoldt traces of one trace-class operator.

The following boundaries are mandatory.

1. The bouquet is not the zeta function of the full map $H_p : \mathbb{F}_p^2 \rightarrow \mathbb{F}_p^2$, and is not a Hasse–Weil local factor.
2. The certified disk is a lower bound for the origin analytic radius. No exact radius, continuation across its boundary, or natural boundary is proved.
3. The odd-prime theorem gives primitive divisors only at odd prime orders. It does not give an eventual composite-order Zsigmondy theorem or squarefreeness.
4. The exact anchor is c_ℓ in the Euler exponent, primitive-cycle ledger, and logarithmic coefficient. The raw coefficient q_ℓ can be larger.
5. The global operator U is noncompact; the locally finite block trace ledger is not a Hilbert-space trace.

Thus:

Gate	Status	First missing object
A	false/open	canonical intrinsic global physical determinant
B	false/open	time-oriented scattering or unitary completion
C	false/open	self-adjoint generator and intrinsic $T \log T$
D	false/open	signed von-Mangoldt prime-power operator trace
E	false/open	completed-zeta divisor equality

No Hilbert–Pólya operator, self-adjoint generator, Riemann-zero spectral identification, completed-zeta divisor equality, or proof of the Riemann Hypothesis follows.

9 Executable protocol

The publication artifact performs finite exact reproduction and provenance checks only:

1. it hashes the locked reversibility, divisibility, finite diagnostic, RH-362, RH-364, and four-volume inputs;
2. it checks the exact midpoint identities through order twelve and an extended test through order sixteen;
3. it checks the quadratic height step and both closed bounds;

4. it factors $a_1, \dots, a_{12}$, verifies each factor's first modular return, and reconstructs c_n , T_n , and q_n ;
5. it checks the prime-order anchors at orders 3, 5, 7, 11, including the raw-coefficient mismatch at orders seven and eleven;
6. it validates the strict analytic-majorant domain, result schema, fixed-member archive, five Gate values, and fifteen forbidden macro claims.

The finite factor rows are not evidence for an all-large-order primitive divisor law. The all-order conclusions are the analytic theorems proved in sections 2–6.

10 Conclusion

The marked return bouquet has more structure than its original formal definition suggested. Reversibility halves the orbit depth of the gcd problem, and the escaping midpoint height pays an explicit all-order fixed-point budget. This produces a genuine zero-free analytic germ without inserting prime weights or changing local clocks.

The same argument also exposes the boundary. Prime order forces new divisors because the only possible earlier rank is one, but composite order does not. The scalar Euler product converges even though the most immediate global cycle operator is noncompact. The shortest new arithmetic problem is therefore a genuine composite-order primitive-divisor theorem; the shortest operator problem is an intrinsic compact globalization rather than another marked direct sum. The original physical same-clock route remains open and unchanged.

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